
Session 5 | Probability & Statistics form Modelling

Exercise sheet

Short questions to work through together at the end of the session. Each one reactivates an idea from the slides and then pushes it a little further: give the number, then say what it means and when it would break down.

Exercise 1 *Recognising a distribution, and its signature.*

For each situation, name the distribution and give its mean and variance.

- (a) The number of heads in 12 tosses of a fair coin.
 - (b) The number of typos on a page, averaging 2 per page.
 - (c) A person's height: discrete or continuous? Why is a Normal a natural model for it (what does the central limit theorem have to say)?
 - (d) Which of these three distributions has its variance *equal* to its mean, and why is that a fingerprint of the Poisson?
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Exercise 2 *A density is not a probability.*

Let X be a continuous random variable.

- (a) What is $\mathbb{P}(X = 5)$ for a single point?
 - (b) Can a probability density $f(x)$ ever exceed 1?
 - (c) For $X \sim \text{Uniform}(0, 2)$ the density is 0.5 on $[0, 2]$. A student reads this as “a 50% chance”. What is wrong with that reading, and what *is* the 50% statement you can make?
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Exercise 3 *Expectation and variance, and what they capture.*

A random variable takes the values 0, 1, 2 with probabilities 0.2, 0.5, 0.3.

- (a) Compute $\mathbb{E}[X]$.
 - (b) Compute $\mathbb{E}[X^2]$, and hence $\text{Var}(X)$.
 - (c) Give $\text{SD}(X)$, and say in words what it measures here.
 - (d) A different variable has the *same* mean but a larger variance. Without recomputing, how does its distribution differ, and is its standard deviation larger or smaller?
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Exercise 4 *The rules, and why they hold.*

Suppose $\text{Var}(X) = 4$ and $\mathbb{E}[X] = 5$.

- (a) Compute $\text{Var}(2X - 1)$ and $\mathbb{E}[3X + 2]$.
 - (b) If Y is independent of X with $\text{Var}(Y) = 9$, what is $\text{Var}(X + Y)$?
 - (c) Why does the shift (-1) leave the variance unchanged while the scale (2) does not? And why was *independence* needed in part (b)?
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Exercise 5 *The z -score compares across different scales.*

Test scores follow $\mathcal{N}(100, 15^2)$.

- (a) What is the z -score of a score of 115?
 - (b) What fraction of people score above 130?
 - (c) Between which two scores does the middle 95% fall?
 - (d) Two students are each 1 standard deviation above the mean, but on tests with *different* standard deviations. Do they share the same raw score? The same percentile? Explain.
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Exercise 6 *The central limit theorem, and what actually shrinks.*

A measurement has population standard deviation $\sigma = 12$.

- (a) What is the standard error of the mean of $n = 36$ measurements?
 - (b) How many measurements bring the standard error down to 1?
 - (c) As n grows, which shrinks: the population standard deviation σ , the standard error, or both? Explain the difference.
 - (d) Why is it striking that the sample mean is approximately Normal *even when a single measurement is very skewed*?
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Exercise 7 *Estimator versus estimate.*

You collect the data $\{2, 4, 6, 8, 10\}$.

- (a) Compute the sample mean $\bar{x}$ and the sample variance s^2 (dividing by $n - 1$).
- (b) Why divide by $n - 1$ rather than n ? Answer in terms of degrees of freedom.
- (c) Your estimate here is $\bar{x} = 6$, yet the true mean could differ. What, then, does it mean to call $\bar{x}$ an *unbiased* estimator?

Exercise 8 *Reading a confidence interval.*

From $n = 25$ measurements with known $\sigma = 15$ you obtain $\bar{x} = 100$.

- (a) Compute the standard error and the 95% confidence interval $(\bar{x} \pm 1.96 \sigma / \sqrt{n})$.
- (b) How does the width change if n rises to 100?
- (c) “95% confidence” is a statement about the *procedure*, not this one interval. Explain the difference in one sentence.
- (d) Would a 99% interval be wider or narrower, and why does more confidence cost width?

Exercise 9 *A hypothesis test and its limits.*

A task has a null mean of 50 with $\sigma = 10$. A group of $n = 100$ scores $\bar{x} = 52$. Test $H_0 : \mu = 50$ against $H_1 : \mu \neq 50$ at $\alpha = 0.05$.

- (a) Compute the standard error and the test statistic z .
- (b) Do you reject H_0 ? (Two-tailed critical value 1.96.)
- (c) A significant result: does it mean the effect is *large*? What role does n play?
- (d) Suppose instead you had *failed* to reject H_0 . Would that prove $\mu = 50$? Explain.

Exercise 10 *The two errors, and the trade-off between them.*

- (a) Define a Type I and a Type II error in one line each.
- (b) If a test uses $\alpha = 0.05$, what is its Type I error rate?
- (c) Name two ways to increase the power of a test.
- (d) If you tighten α from 0.05 to 0.01 to avoid false alarms, what happens to the Type II error rate and the power? Why is this a trade-off?

Exercise 11 *Why the base rate dominates.*

A disease affects 5% of people. A test is positive for 80% of those who have it (sensitivity) and for 10% of those who do not (false-positive rate). You test positive.

- (a) Compute the total probability of a positive test, $\mathbb{P}(+)$.
- (b) Compute $\mathbb{P}(\text{disease} | +)$.
- (c) Reconcile the two numbers: the test is “80% accurate”, yet a positive gives only about a 30% chance of disease. How can both be true?

- (d) Which would raise $\mathbb{P}(\text{disease} | +)$ more: a somewhat better test, or first screening down to a higher-risk group? Explain briefly.
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Exercise 12 *Bayes, and the strength of the evidence.*

- (a) Write Bayes' theorem for $\mathbb{P}(A | B)$.
- (b) Given $\mathbb{P}(B | A) = 0.8$, $\mathbb{P}(A) = 0.25$ and $\mathbb{P}(B) = 0.4$, compute $\mathbb{P}(A | B)$.
- (c) The prior was 0.25 and the posterior is 0.5: by what factor did observing B update the belief, and where does that factor come from (relate it to $\mathbb{P}(B | A)$ and $\mathbb{P}(B)$)?
- (d) In the disease example, would a *higher* base rate raise or lower $\mathbb{P}(\text{disease} | +)$?
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Exercise 13 *A glimpse ahead: the modelling pipeline.*

- (a) In one sentence, what does a *cognitive model* add beyond describing the data?
- (b) In the pipeline data $\rightarrow$ model $\rightarrow$ fit $\rightarrow$ uncertainty $\rightarrow$ compare, which step does maximum likelihood carry out?
- (c) Name one model parameter that could stand for something psychological.
- (d) Which tools from today would you reach for at the *uncertainty* step?
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Exercise 14 *MLE, MAP, and the role of the prior.*

- (a) MLE maximises the _____; MAP maximises the _____ (likelihood $\times$ prior).
- (b) Under what condition do MLE and MAP give the same estimate?
- (c) With very little data, which of the two is pulled more strongly by the prior?
- (d) Give one situation where deliberately choosing a *non-flat* prior is an advantage, and say what it protects against.